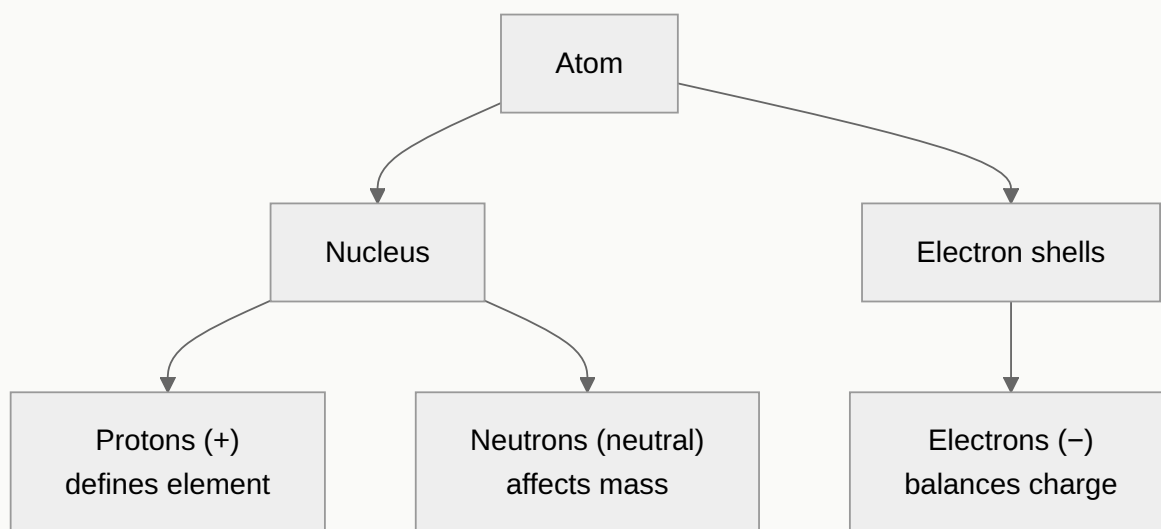


Atom

An atom is the smallest unit of an element that retains the chemical identity of that element — the basic building block described in Dalton's atomic theory (1803), which proposed that matter is made of indivisible atoms unique to each element. Modern atomic theory refines "indivisible": every atom is built from three subatomic particles. Protons (positive charge, ~ 1 atomic mass unit) and neutrons (no charge, ~ 1 amu) occupy a dense central nucleus, while electrons (negative charge, negligible mass) surround the nucleus in energy levels, or shells. The number of protons, called the atomic number, is what defines the element — every atom with 6 protons is carbon, no matter what else changes about it.

Worked example. Consider carbon-12, the standard reference isotope used to define the atomic mass unit. Its atomic number is 6, so it has 6 protons. Its mass number — the total count of protons plus neutrons — is 12, so it has 6 neutrons. A neutral atom always carries an equal number of electrons and protons, so carbon-12 also has 6 electrons balancing the nuclear charge. Note that the proton count alone tells you the element; the neutron count only affects the atom's mass, not its chemical behavior.

Problem-solving application. Most atomic-structure problems ask you to reconstruct particle counts from two numbers: the atomic number and the mass number. Given an atom with atomic number 17 and mass number 35: protons = 17 (read directly from the atomic number), neutrons = $35 - 17 = 18$ (mass number minus proton count), and electrons = 17, since a neutral atom balances proton and electron counts exactly. This three-step bookkeeping — protons from atomic number, neutrons by subtraction, electrons by charge balance — applies to any atom on the periodic table and is the foundation for everything that follows in later sections: how atoms of the same element can differ in mass, how they can gain or lose charge, and how their electron arrangements drive chemical bonding.



A schematic breakdown of atomic structure: the nucleus holds protons and neutrons, while electrons occupy surrounding shells.